

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
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Section / Batch:	2023	Date:	27/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q47. Closest Pair – Points on a Grid

Given the set of points $\{(0,1), (1,2), (2,1), (3,3), (4,4), (2,2)\}$, apply the Closest Pair Brute Force algorithm. Clearly interpret the problem and justify the chosen approach. Compute the distances between every possible pair and show the calculations. Identify the closest pair and count the number of comparisons performed. Analyze the time complexity of the brute force method. Verify the correctness of the result and interpret its significance in grid-based geometry problems.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q47. Closest Pair – Points on a Grid

Given Points

{(0,1), (1,2), (2,1), (3,3), (4,4), (2,2)}

Problem Interpretation

The Closest Pair Problem asks us to find the two points whose Euclidean distance is the smallest among all possible pairs of points. Since no information about the distribution of points is given, the Brute Force approach is chosen because it checks every possible pair of points and guarantees the correct answer.

The Euclidean distance between two points is calculated using the formula:

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

Method: Examining All Possible Pairs

Let the points be:

P1 = (0,1)

P2 = (1,2)

P3 = (2,1)

P4 = (3,3)

P5 = (4,4)

P6 = (2,2)

For n = 6 points,

Total number of pairs = $n(n - 1)/2$

= $6 \times 5 / 2$

= 15 pairs

The brute-force algorithm computes the Euclidean distance for each of the 15 pairs and keeps track of the smallest distance.

Distance Calculations

Pair	Distance Calculation	Distance
P1–P2	$\sqrt{[(1-0)^2 + (2-1)^2]} = \sqrt{2}$	1.414
P1–P3	$\sqrt{[(2-0)^2 + (1-1)^2]} = \sqrt{4}$	2
P1–P4	$\sqrt{[(3-0)^2 + (3-1)^2]} = \sqrt{13}$	3.606
P1–P5	$\sqrt{[(4-0)^2 + (4-1)^2]} = \sqrt{25}$	5
P1–P6	$\sqrt{[(2-0)^2 + (2-1)^2]} = \sqrt{5}$	2.236
P2–P3	$\sqrt{[(2-1)^2 + (1-2)^2]} = \sqrt{2}$	1.414

P2–P4	$\sqrt{[(3-1)^2 + (3-2)^2]} = \sqrt{5}$	2.236
P2–P5	$\sqrt{[(4-1)^2 + (4-2)^2]} = \sqrt{13}$	3.606
P2–P6	$\sqrt{[(2-1)^2 + (2-2)^2]} = \sqrt{1}$	1
P3–P4	$\sqrt{[(3-2)^2 + (3-1)^2]} = \sqrt{5}$	2.236
P3–P5	$\sqrt{[(4-2)^2 + (4-1)^2]} = \sqrt{13}$	3.606
P3–P6	$\sqrt{[(2-2)^2 + (2-1)^2]} = \sqrt{1}$	1
P4–P5	$\sqrt{[(4-3)^2 + (4-3)^2]} = \sqrt{2}$	1.414
P4–P6	$\sqrt{[(2-3)^2 + (2-3)^2]} = \sqrt{2}$	1.414
P5–P6	$\sqrt{[(2-4)^2 + (2-4)^2]} = \sqrt{8}$	2.828

Identifying the Closest Pair

After checking all 15 pairs, the smallest distance obtained is 1.

The closest pairs are:

(1,2) and (2,2)

(2,1) and (2,2)

Minimum Distance = 1 unit

Comparisons Performed

The brute-force algorithm compares every possible pair.

Total comparisons = $n(n - 1)/2$
 $= 6 \times 5 / 2$
 $= 15$ comparisons.

Time Complexity Analysis

Approach	Description	Complexity
Brute Force	Computes distance for every pair	$O(n^2)$
Space Complexity	Uses only a few variables	$O(1)$

For $n = 6$:

- Total pairs evaluated = 15.
- Running time grows proportional to n^2 because every pair must be checked.

Verification of Correctness

The brute-force algorithm examines every possible pair of points exactly once. Since no pair is skipped, the smallest distance found is guaranteed to be the actual minimum.

The minimum distance of 1 unit occurs between:

- (1,2) and (2,2)
- (2,1) and (2,2)

Hence, the result is correct.

Interpretation of Efficiency

- The brute-force approach is simple, exhaustive, and guarantees the correct answer because it checks every possible pair.
- It works well for small datasets where the number of points is limited.
- However, as the number of points increases, the number of comparisons grows rapidly according to $O(n^2)$, making it inefficient for very large datasets.
- More efficient algorithms such as the Divide and Conquer Closest Pair Algorithm solve the problem in $O(n \log n)$ time and are preferred for large datasets.

Conclusion

The Brute Force Closest Pair algorithm evaluated all 15 possible pairs of the given six points. The minimum Euclidean distance obtained was 1 unit, occurring between the pairs ((1,2), (2,2)) and ((2,1), (2,2)). Since every possible pair was examined, the result is guaranteed to be correct. Although the brute-force method has $O(n^2)$ time complexity and becomes inefficient for large datasets, it is easy to understand, simple to implement, and useful for verifying the correctness of more efficient algorithms.